

Pipeline Hazards & Resolution

Q. Define Pipeline Hazards. Classify and explain the three main types of pipeline hazards (Structural, Data, and Control) and briefly mention methods to handle them.

[DEF] Definition to Remember

A Pipeline Hazard occurs when the next instruction in a pipeline cannot execute in its designated clock cycle. This stalls the pipeline (creates a "bubble"), preventing it from achieving maximum instruction throughput.

1. Structural Hazards

Occurs when two instructions in the pipeline need the same hardware resource at the exact same time (Resource Conflict).

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2. Data Hazards

Occurs due to Data Dependency, meaning an instruction needs the result of a previous instruction that hasn't finished executing yet.

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3. Control (Branch) Hazards

Occurs due to Branch instructions (Jump, If-Else). The pipeline fetches instructions sequentially. When a branch is encountered, the target address isn't known until the execute stage. Meanwhile, wrong instructions may have been fetched.

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[MUST-WRITE] Must-Write Points (for 10 marks)

1. Pipeline Hazard: A situation preventing the next instruction from executing in its proper cycle.

2. Three Types: Structural, Data, and Control Hazards.

3. Structural Hazard: Resource conflict (two instructions need the same hardware). Fixed by duplicating resources (Instruction vs Data cache).

4. Data Hazard: Data dependency (Instruction needs data from an unfinished prior instruction).

5. Data Hazards are fixed via Data Forwarding or compiler scheduling.

6. Control Hazard: Caused by branch/jump instructions fetching the wrong sequential path.

7. Control Hazards are fixed via Branch Prediction, flushing, or delayed branching.

[RECALL] Quick Recall

Pipeline Hazards -> Structural (Hardware conflict -> fix: duplicate resources) -> Data (Dependency -> fix: Data Forwarding/Bypass) -> Control (Branching -> fix: Branch Prediction/Flushing)